

Predicting Crime Categories Using Machine Learning

Abstract

This project utilizes a machine learning approach to predict the category of crimes based on various features extracted from a public dataset of crime data in Los Angeles, spanning from 2020 to the present. The dataset includes information about crime types, victim demographics, weapon descriptions, time, and location.

The pipeline integrates preprocessing steps, feature engineering, and a powerful machine learning algorithm, LightGBM, to perform multiclass classification. Key characteristics of the project are:

- **Dataset Cleaning and Preparation:** The dataset was cleaned by handling missing values, removing outliers (e.g., invalid victim ages), and transforming categorical variables into numeric format using dummy encoding.
- **Feature Engineering:** New features were created, such as:
 - **Time Category** (e.g., Morning, Afternoon, Evening, Night) derived from the time of occurrence.
 - **Season** (e.g., Winter, Spring) extracted from the date of the crime.
 - **Macro Area** mapped from specific zones of Los Angeles.
- **Model Selection:** The LightGBM classifier was chosen for its efficiency and ability to handle multiclass classification tasks.
- **Performance Evaluation:** The model was evaluated using cross-validation, a confusion matrix, and classification metrics such as precision, recall, and F1-score to ensure robust performance.

This model aims to provide insights into crime patterns and support resource allocation for crime prevention. It demonstrates the potential of machine learning in analyzing and understanding real-world crime data.